SPHERICAL COW PHILOSOPHY: A METHODOLOGICAL FRAMEWORK FOR MATHEMATICAL DISCOVERY

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Abstract. We present *spherical cow philosophy*—a methodological framework for mathematical discovery that complements classical proof-by-contradiction by accepting empirical constitutional constraints as foundational and deriving their forced consequences. Named after the physics heuristic of modelling complex systems through essential simplification, the approach inverts the usual hierarchy: instead of deriving special cases from universal axioms, we identify structural constraints that are empirically universal within a domain, derive what they force rigorously, and validate predictions across independent domains.

The framework addresses a specific situation: when classical methods encounter persistent structural barriers—such as the parity barrier that keeps sieve methods from twin primes, or the level $1/2$ of the Bombieri–Vinogradov theorem (1965), which for general moduli has been exceeded only in restricted settings—the productive path may be to accept a constitutional constraint (here, $p \equiv 2 \pmod 3$ for all twin primes $p > 3$) as the natural entry point and derive its arithmetic consequences, rather than to approach the barrier as an obstacle for more powerful general methods.

Applied to twin primes, this methodology motivated the cascade moduli framework of the companion paper [19], which conjectures a level of distribution $\theta_W = 5/8$ for the moduli $q = 3^K$, and a proposed analogy with information theory [21]. We show that the consequences claimed for the twin-prime constitution in the February version are not forced by it, and we use the case study to sharpen the methodology: a consequence that depends on a further choice is not forced, and a prediction counts only if it is fixed before the comparison and could fail.

We give criteria for when the framework applies, contrast it with classical approaches, and distinguish it from circular or post-hoc reasoning. Logical status labels are attached to every claim.

Revision note (November 2026). The "five forced consequences" of the twin-prime constitution are no longer claimed to be forced; Section 4.2 explains, item by item, why they are not, and $\theta_W = 5/8$ is described as the conjecture it is. The statement that a level of distribution $\theta > 1/2$ would yield twin primes is withdrawn: because of the parity barrier, no level of distribution does so. The Sophie Germain "computational confirmation" is withdrawn, the comparison with Shannon's AEP is corrected (the typical set is a vanishing fraction of the space, the opposite of the θ_k pattern), and the degraded-channel remark is reversed. The history of the $\theta = 1/2$ barrier is corrected (Bombieri–Vinogradov, 1965; levels above $1/2$ are known for well-factorable weights), as are the Bombieri and Pascadi citations, and the unverifiable Clark quotation is removed. Smaller errors are fixed: the 78-year gap, the hypotheses of Gödel's theorem, the physics analogy and dangling cross-references; the programme is now described as nine papers.

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1. Introduction: The Physics Analogy

1.1. **The original spherical cow.** When physicists model heat transfer from a cow, they make a radical simplification: assume a spherical cow. Strip away legs, spots, tail, digestive complexity—keep only mass distribution and thermal properties. The point of the joke is that a deliberately crude description can be the first step towards an accurate one. The spherical cow succeeds because it knows what to *ignore*. Domain-specific details obscure the essential physics. By ignoring the irrelevant, physicists isolate what matters, solve the simplified problem, and add complexity back strategically where it is needed.

1.2. **The mathematical parallel.** Classical mathematics has three primary responses to a hard problem:

(1) Assume the statement false, derive consequences, seek a contradiction.
(2) Build from axioms: prove the statement from more fundamental principles.
(3) Exhaust known techniques until something works.

These work brilliantly when the problem is provable from accepted axioms. When they encounter a persistent structural barrier, the barrier itself is information. It may be signalling that the productive formulation has not yet been identified—not that the problem is harder than expected.

Spherical cow philosophy provides a complementary response: instead of asking "how do we overcome this barrier?", ask "what constraint is this barrier telling us is constitutional?" Accept that constraint as the productive entry point. Derive what it forces. Validate the consequences.

1.3. **Position in the programme.** This paper is p03 of a nine-paper programme. Its role is methodological: it explains the reasoning behind the programme's arithmetic papers and, in this revision, uses them as a case study of where that reasoning holds and where it fails. The other eight papers, with the revised status of their main claims, are:

- [17] (p01) defines Constitutional Forcing; the formula $\theta_k = (2^k - k)/2^k$ rests on a counting postulate, not a theorem.
- [18] (p02) gives the history of Khayyam's Triangle and its mod-3 fractal; the claimed link between the fractal and θ_k is withdrawn.
- [19] (p04) conjectures $\theta_W = 5/8$ for cascade moduli $q = 3^K$, using the arithmetic of the Eisenstein integers $\mathbb{Z}[\omega]$.
- [20] (p05) states the formula $\theta_k = (2^k - k)/2^k$ as a postulate.
- [21] (p06) proposes an analogy between the arithmetic programme and Shannon's theorems in information theory.
- [22] (p07) treats Condition W3, the absence of Siegel zeros for characters modulo 3^K; it is true, with a one-line proof from the cyclicity of $(\mathbb{Z}/3^K\mathbb{Z})^\times$, and it affects no level of distribution.
- [23] (p08) discusses the Lemke Oliver–Soundararajan bias in consecutive primes; its claimed ternary explanation is withdrawn.
- [24] (p09) treats the conjugate symmetry of the real discrete Fourier transform.

The present paper explains *why* that programme is structured the way it is, and what methodological principle motivated it.

1.4. **What this paper claims and does not claim.** We attach logical-status labels to every significant claim throughout. Here we state the top-level ones explicitly.

[Claimed] Accepting a proved structural fact as the *productive entry point* to a hard problem is a legitimate methodological move. In the twin-prime case study, the entry point $p \equiv 2 \pmod 3$ motivated the cascade moduli framework of [19], whose central statement $\theta_W = 5/8$ is a conjecture.

[Claimed] The constitution $p \equiv 2 \pmod 3$ is a theorem (provable in two lines), not an axiom. Accepting a proved fact as the *productive entry point*—rather than a consequence to be derived from something deeper—is the methodological move this paper advocates.

[Claimed] The framework can be made falsifiable, provided every prediction, including the value of every parameter, is fixed before it is compared with known values. The $k = 4$ prediction (Remark 5.5) is open but does not yet meet this standard. The Sophie Germain example of the February version is not a test (Remark 5.1).

[Not claimed] That constitutional forcing, or any level of distribution, proves twin prime infinitude. Even level $\theta = 1$ gives only bounded gaps by GPY-type sieves, and the parity barrier blocks gap 2 (Example 2.3).

[Not claimed] That twin prime infinitude is independent of ZFC in the technical Gödel–Cohen sense. We make no claim about independence. The methodology applies whenever constitutional structure is the productive entry point, regardless of the problem's axiomatic status.

2. The Classical Toolkit and Its Limits

2.1. **When classical methods succeed.** Proof by contradiction is among the most powerful tools in mathematics.

Example 2.1 (Irrationality of $\sqrt{2}$)**.** Assume $\sqrt{2} = p/q$ in lowest terms. Then $2q^2 = p^2$, forcing p even; writing $p = 2k$ gives $q^2 = 2k^2$, forcing q even—contradicting lowest terms. Therefore $\sqrt{2}$ is irrational.

This works because the statement is provable from the axioms of arithmetic: the assumption of rationality produces a genuine logical contradiction.

Example 2.2 (Infinitude of primes, Euclid)**.** Suppose there are finitely many primes $p_1, \ldots, p_n$. The number $N = p_1 \cdots p_n + 1$ is divisible by none of them, so it has a prime factor not in the list—a contradiction.

Both examples share a structure: *the assumption of falsehood leads directly to a contradiction derivable from accepted axioms.* Classical proof by contradiction works exactly when this structure is present.

2.2. **Where structural barriers arise.** Some problems resist sustained classical effort not because they are unprovable, but because the productive formulation has not yet been identified. The barrier is information: it is telling us something about the structure of the problem.

Example 2.3 (The $\theta = 1/2$ barrier and the parity barrier for twin primes)**.** The Goldston–Pintz–Yıldırım theorem [10] shows that a level of distribution $\theta > 1/2$ for primes in arithmetic progressions would imply the existence of infinitely many prime pairs with bounded gaps. The Bombieri–Vinogradov theorem [6, 9] gives $\theta = 1/2$ on average over the moduli $q \leq x^{1/2}(\log x)^{-B}$; it is not a statement about each modulus, and even the Generalised Riemann Hypothesis gives the expected asymptotic for a single modulus only for $q \leq x^{1/2-\varepsilon}$. Zhang [11] obtained a level $1/2 + \delta$, for a small $\delta > 0$, for smooth squarefree moduli, which sufficed for a variant of GPY and proved bounded gaps. Maynard and Tao [12] then obtained bounded gaps from level $1/2$ (indeed from any positive level) and improved the gap bound substantially. But gap 2 (twin primes specifically) remains out of reach, and not for want of a higher level: under the Elliott–Halberstam conjecture (level 1) these methods give gaps at most 12 [12], under its generalised form at most 6 [16], and Selberg's parity barrier prevents such sieve methods from reaching gap 2 at any level of distribution.

The level $1/2$ for general moduli reflects the large sieve inequality, but it is not absolute: for a fixed residue class and suitably well-factorable sieve weights, the levels $4/7$ [7], $3/5$ [13], $66/107$ [14] and $5/8$ [15] are known. Structure specific to a single prime is also exploited classically: for prime-power moduli $q = p^N$ the expected asymptotic is known for $q \leq x^{3/8-\varepsilon}$

(Barban–Linnik–Tshudakov [5]), extended to 2/5 (Gallagher [8]) and 5/12 (Huxley), with lower bounds of the correct order up to $x^{0.4736}$ (Banks–Shparlinski); see [4] for an account. All of these prime-power exponents are below 1/2.

The spherical cow response to Example 2.3 is not to work harder on the general barrier. It is to ask: *what constitutional structure do twin primes carry that distinguishes them from the general case?*

The answer—$p \equiv 2 \pmod 3$ for all twin primes $p > 3$—is a two-line theorem. In this programme, accepting it as the entry point led to the cascade moduli $q = 3^K$, to the Eisenstein integers, and to the conjecture $\theta_W = 5/8$ of [19]. Section 4.2 explains why none of these is forced by the constitution. Because of the parity barrier, no improvement in the level of distribution would by itself settle the twin prime conjecture; the case study illustrates the method, not a route to that conjecture.

2.3. **The structural limitation of the classical approach.** Many classical methods seek *universal* patterns: results that hold for all moduli, all primes, all arithmetic progressions. This universality is their strength and their limitation. When a problem has constitutional structure that singles out a specific prime (here, 3), universal methods may ignore that structure.

Spherical cow philosophy embraces the specific structure from the outset. It trades universality for depth. The tools it brings in—the hexagonal lattice of $\mathbb{Z}[\omega]$, the CM field $\mathbb{Q}(\omega)$, character sums over the cyclic group $(\mathbb{Z}/3^K\mathbb{Z})^\times$—are not invisible to classical methods: the prime-power-modulus literature cited in Example 2.3 uses the cyclic and p-adic structure of $(\mathbb{Z}/p^N\mathbb{Z})^\times$ systematically. What the methodology changes is the order in which questions are asked, not the tools available.

3. Spherical Cow Philosophy: The Framework

3.1. **The inversion.** Classical hierarchy:

$$\text{Universal Axioms} \longrightarrow \text{General Theorems} \longrightarrow \text{Special Cases} \longrightarrow [\text{barrier}]$$

Spherical cow hierarchy:

$$\begin{aligned}\text{Empirical Constitution} &\longrightarrow \text{Forced Consequences} \\ &\longrightarrow \text{Cross-Domain Validation} \longrightarrow \text{Falsifiable Predictions}\end{aligned}$$

The key difference: we do not try to prove the constitution from something "more basic." We show the constitution generates observed structure across multiple independent domains and makes predictions that can be confirmed or refuted.

3.2. **The five steps.**

Definition 3.1 (Spherical Cow Methodology)**.** Given a domain with a persistent structural barrier, the spherical cow methodology proceeds as follows.

Step 1: Identify the empirical constitution. Find a constraint that is:

(a) *Verifiable by direct computation*—not an assumption;
(b) *Universal within the domain*—all instances satisfy it;
(c) *Simple to state*—not complex conditional logic.

Step 2: Derive forced consequences. Using rigorous mathematics, determine what the constitutional constraint *forces*. Do not add hypotheses; derive consequences. A consequence that depends on a further choice (of a family of moduli, of a number of levels, of a counting rule) is not forced; record the choice as a hypothesis.
Step 3: Validate within the domain. Check that the forced consequences match domain observations, both computationally and analytically where possible.
Step 4: Validate across independent domains. Determine whether the same constitutional structure explains phenomena in genuinely independent fields. Independent confirmation strengthens the case that the constitution is fundamental, not accidental.
Step 5: Generate falsifiable predictions. Identify consequences of the constitution that have not yet been observed, and check whether they hold or fail. A prediction must be fixed, including the value of every parameter, before it is compared with known values, and it must be capable of failing. A framework that makes and confirms such predictions—and identifies genuine non-instances—is more than decorative.

3.3. **The spherical cow principle.**

Principle 3.2 (Spherical Cow). What to ignore: domain-specific complexity that obscures constitutional structure.
What to keep: the constitutional constraint that generates forced consequences.
How to validate: cross-domain prediction and confirmed non-instances.

The physics analogy is precise. A physicist ignores the cow's legs and spots to isolate thermal properties. A mathematician ignores the full generality of all moduli to isolate the constitutional property of twin primes ($p \equiv 2 \pmod 3$) and to ask what structure it suggests.

4. Case Study: Twin Primes

4.1. **The constitution.**

Proposition 4.1 (Twin Prime Constitution)**.** *Every twin prime $p > 3$ satisfies $p \equiv 2 \pmod 3$.*

Proof. Among any three consecutive integers, exactly one is divisible by 3. For the pair $(p, p+2)$: if $p \equiv 0 \pmod 3$ then $3 \mid p$, so p is not prime (for $p > 3$). If $p \equiv 1 \pmod 3$ then $p + 2 \equiv 0 \pmod 3$, so $3 \mid p + 2$, hence $p + 2$ is not prime (for $p + 2 > 3$). Therefore $p \equiv 2 \pmod 3$. □

Remark 4.2. Proposition 4.1 is a theorem, not an axiom. The methodological move is not to assume it but to *accept it as the productive entry point*—to ask what arithmetic structure it forces, rather than treating it as a consequence to be recovered from something more general.

4.2. **The proposed consequences, and why they are not forced.**

Given Proposition 4.1, the spherical cow methodology asks: what does $p \equiv 2 \pmod 3$ force? The February version listed five "forced consequences". We record each, with what is actually true.

(1) **Cascade structure.** The constitution is a condition modulo 3, so it suggests the moduli $q = 3^K$—the *cascade moduli*. It does not force them, and it imposes nothing modulo 9 or 27 beyond $p \equiv 2 \pmod 3$: every lift of 2 mod 3 to modulus 3^K keeps both p and $p + 2$ prime to 3. (That conditions modulo 3^K and modulo a prime $\ell \neq 3$ are independent is the Chinese Remainder Theorem, whose hypothesis $\gcd(3^K, \ell) = 1$ is not a consequence of the constitution.)

(2) **Eisenstein geometry.** The prime 3 ramifies in the Eisenstein integers $\mathbb{Z}[\omega]$ as $(3) = (1-\omega)^2$, and a prime $\ell > 3$ is *inert* if $\ell \equiv 2 \pmod 3$ and *splits* if $\ell \equiv 1 \pmod 3$. So for a twin pair $(p, p+2)$ with $p > 3$, p is inert and $p+2$ splits. This is true, but it is a restatement of the constitution, not a new constraint.

(3) **Three-level filtration** (withdrawn). The February version claimed a filtration of $\mathbb{Z}[\omega]/(3^K)$ with exactly three independent levels. Since $(3^K) = (1-\omega)^{2K}$, the $(1-\omega)$-adic filtration has $2K$ steps, each with quotient $\mathbb{F}_3$. "Triply-well-factorable" is a property imposed on sieve weights for general moduli [14, 15]; it is not forced by $\mathbb{Z}[\omega]$.

(4) **Level of distribution** $\theta_W = 5/8$ (conjecture). The February version described a ceiling $\theta \le 2/3$ from three-fold Cauchy–Schwarz which cyclic-group cancellation "improves" to $5/8$. Since $5/8 < 2/3$ that would be a weaker exponent, not an improvement, and the argument rested on a pointwise Kloosterman bound $q^{3/8}$ for $q = 3^K$ that is false: Weil's bound $2q^{1/2}$ is essentially sharp for these moduli (for each $3 \le K \le 7$ some $|S(1, b; 3^K)|$ exceeds $1.99 \cdot 3^{K/2}$). As stated in [19], the estimate summed over the moduli $3^K \le x^\theta$ holds for every fixed $\theta < 1$ (by the trivial bound for $3^K > (\log x)^{A+1}$ and the Siegel–Walfisz theorem for the $O(\log\log x)$ smaller moduli), so it cannot single out $5/8$; the per-modulus version would go beyond what GRH gives. $\theta_W = 5/8$ is therefore a conjecture. Condition W3 [22] is true, with a one-line proof from the cyclicity of $(\mathbb{Z}/3^K\mathbb{Z})^\times$, but the summed estimate does not need it and it makes nothing unconditional.

(5) **Combinatorial origin** (withdrawn). The value $5/8 = (2^3-3)/2^3$ is the configuration density at depth $k = 3$ under the counting postulate of [17, 20]. It is not derived: if "invalid" means that some constraint fails, $2^3 - 1 = 7$ of the 8 configurations are invalid. The depth 3 is not supported by items (1) or (3). Nor does the value trace to the mod-3 Sierpiński structure of Khayyam's Triangle: in the first 3^K rows the proportion of entries not divisible by 3 is $2^{K+1}/(3^K + 1)$, that is $1, 4/5, 4/7, 16/41, \ldots \to 0$, whereas $(2^K - K)/2^K \to 1$.

None of the five is forced by Proposition 4.1 alone. Item (2) restates it; items (1), (3) and (5) rest on choices—of moduli, of a number of levels, of a counting rule—that the constitution does not determine; and item (4) is a conjecture. This is the main methodological lesson of the case study: Step 2 of Definition 3.1 must be carried out with the rigour of a proof, and a consequence that depends on a choice is a hypothesis, not a forced consequence.

4.3. **The GPY connection.** The Goldston–Pintz–Yıldırım method [10] uses a level of distribution $\theta > 1/2$, for sieve weights supported on squarefree moduli, to obtain bounded gaps between primes. No level of distribution, even $\theta = 1$, yields twin primes by these methods (Example 2.3). The moduli 3^K with $K \ge 2$ are not squarefree, so they do not occur among the moduli used by GPY-type sieves.

[Withdrawn] The February version described the extension of $\theta_W = 5/8$ to a family of moduli broad enough to serve as GPY input as "the principal open step" towards twin primes, with one avenue through Pascadi's result [15]. That step would not lead to twin primes. Pascadi's theorem—equidistribution of smooth numbers, and of primes with triply-well-factorable weights, in arithmetic progressions to general moduli up to $x^{5/8-o(1)}$—concerns general moduli, not the moduli 3^K; its agreement with $\theta_3 = 5/8$ is a numerical coincidence.

The spherical cow methodology motivated and structured the arithmetic programme. At present the programme's central arithmetic statement is a conjecture.

4.4. **Cross-domain comparison: Shannon's theorems.** The programme compares the twin-prime case with Shannon's two foundational theorems of 1948 [1]; see [21].

- **Asymptotic Equipartition Property.** Of the $|\mathcal{A}|^n$ sequences of length n from a memoryless source, a typical set of about 2^{nH} sequences carries almost all of the probability; the entropy H is determined by the source distribution.
- **Channel Coding Theorem.** Reliable communication is possible at rates $R < C$ and not at rates $R > C$, where the capacity $C = \max_{p(x)} I(X;Y)$ is determined by the channel.

The comparison is an analogy, and in one respect an inverse one. In the AEP the typical set is a vanishing fraction of the space: $|A_\varepsilon^{(n)}|/|\mathcal{A}|^n \le 2^{-n(\log|\mathcal{A}|-H-\varepsilon)} \to 0$ unless the source is uniform (logarithms to base 2, ε small). In the θ_k scheme the valid fraction $1 - k/2^k$ tends to 1: the constraints remove a vanishing fraction. Moreover the governing constants $H/\log|\mathcal{A}|$ and C take every value in $[0,1]$, so they are not $\theta_1 = 1/2$ [17, Section 3.3]. The two settings are therefore not instances of a single pattern; at most they share the feature that a constant fixed by the structure of the system governs the size of a distinguished subset. The comparison remains a proposed analogy.

Shannon's results (1948) precede the arithmetic conjecture (2026) by 78 years. Chronology alone does not show that neither side was constructed to match the other: the value $5/8$ coincides with Pascadi's exponent for general moduli [15], which predates the companion papers. A comparison is free of post-hoc fitting only if its parameters (here k) are fixed before the constants are compared.

5. Non-Circularity and Falsifiability

5.1. **Why this is not circular.** A potential objection: the framework accepts the constitution, derives consequences, and validates against observations in the same domain. Is this circular?

The answer depends on how the methodology is carried out, and the case study shows both the safeguard and the failure.

First, the constitution is a theorem (Proposition 4.1), not an assumption. Accepting a proved fact as the entry point is not circular; it is a choice of direction.

Second, the methodology is non-circular only if the consequences are genuinely derived. In the case study they were not: the value $5/8$ entered as an input, through the choice of three levels (items (3) and (5) of Section 4.2), and the estimate of [19] as stated holds for every $\theta < 1$, so it cannot single out $5/8$.

Third, a falsifiable prediction must be one that could fail. The Sophie Germain example offered in the February version as a confirmed prediction is not one.

Remark 5.1 (Sophie Germain primes: not a test). For $p > 3$, if p and $2p+1$ are both prime then $p \equiv 2 \pmod 3$—the same single condition as for twin primes. In both cases the partner ($p+2$ or $2p+1$) is an affine function of p, so its residue modulo 3^K is determined by that of p. The February version assigned Sophie Germain pairs depth $k = 2$ and twin primes depth $k = 3$, but their local structure does not distinguish them: modulo every prime $\ell \ge 3$ both exclude exactly two residue classes of p (for $\ell = 3$ the same classes, 0 and 1; for $\ell \ge 5$ different ones, $\{0,-2\}$ against $\{0,-1/2\}$), so their local densities, and their Hardy–Littlewood constants, are equal. Since $\theta_1 = \theta_2 = 1/2$, the "prediction" $\theta = 1/2$ could not fail whichever depth is assigned. Moreover, a level of distribution is a property of primes in arithmetic

progressions, the same input for every admissible tuple, so a "level of distribution of Sophie Germain pairs" is not a defined notion; and no finite computation can confirm an asymptotic level. The example, and its claimed computational confirmation, are withdrawn as evidence.

Remark 5.2. The February version compared the Sophie Germain coupling with a degraded channel, saying that a degraded channel has lower capacity because input–output dependence increases. That is backwards. For a degraded channel $X \to Y \to Z$ the data-processing inequality gives $I(X;Z) \leq I(X;Y)$, so the capacity $\max_{p(x)} I(X;Z)$ is at most $\max_{p(x)} I(X;Y)$ because input–output dependence *decreases*. In any case, by Remark 5.1 the coupling in Sophie Germain pairs is the same as in twin primes, so the comparison distinguishes nothing.

5.2. **Decision criteria: when to use this framework.**

Criterion 5.3 (Apply spherical cow philosophy when:)**.**
(1) **Structural barrier**: Classical methods have encountered a persistent ceiling that is not merely a technical obstacle but appears to reflect the limits of the general approach.
(2) **Identifiable constitution**: A constraint exists that is universal within the domain, verifiable computationally, and simple to state.
(3) **Cross-domain echo**: The same structural pattern appears in independent fields, suggesting the constitution captures something fundamental rather than domain-specific.
(4) **Falsifiable predictions**: The framework generates predictions about cases not yet examined, fixed in advance, which can confirm or refute it.

Criterion 5.4 (Prefer classical methods when:)**.**
(1) The statement is provable from accepted axioms by known techniques.
(2) The problem is single-domain without structural echoes elsewhere.
(3) No universal constitutional constraint is identifiable.
(4) Proof by contradiction successfully finds contradictions.

Remark 5.5 (The $k = 4$ open prediction). The programme proposes, as its next prediction, a prime constellation with $k = 4$ independent modular constraints and governing constant $\theta_4 = (2^4 - 4)/2^4 = 3/4$ (see Table 1 of [17]). In its present form this is not yet a prediction with operational content. A level of distribution is a property of primes in arithmetic progressions, the same input for every admissible tuple, so "the level of distribution of a constellation" would first have to be defined; the four constraints, and their independence, would have to be fixed before any comparison; and since a level of distribution is an asymptotic statement, a finite computation could illustrate it but neither verify nor disprove it. Supplying these is an open problem.

6. Philosophical Foundations

6.1. **Empiricism and rationalism in mathematics.**

Classical mathematics is rationalist: truth derives from axioms and logical deduction. Empirical observations serve as heuristics, not proofs.

Spherical cow philosophy is *methodologically* empiricist, without abandoning rigour. Constitutional constraints are accepted as entry points because they are empirically universal and computationally verifiable. The *derivation* from those constraints remains fully rigorous. The shift is in what counts as the productive starting point, not in what counts as a proof.

The physics analogy is again apt. Physicists accept the invariance of the speed of light in all inertial frames—an empirically supported postulate (Einstein's second postulate)—and rigorously derive the structure of spacetime from it. The derivation (special relativity, $E = mc^2$, time dilation) is mathematically exact; the starting point is empirical. Spherical cow philosophy proposes the same division of labour for mathematics: accept the constitutional constraint empirically, derive its consequences rigorously.

6.2. **Gödel, independence, and the scope of this claim.** Gödel's incompleteness theorems [2] establish that any consistent, effectively axiomatisable formal system containing enough arithmetic has true arithmetical statements unprovable within the system. (Effective axiomatisability is essential: the set of all true arithmetical statements is consistent and complete.) Cohen [3], completing earlier work of Gödel, showed that the continuum hypothesis is independent of ZFC.

We do *not* claim that twin prime infinitude is independent of ZFC. We make no claim about the axiomatic status of any conjecture in this programme.

What we do claim is narrower and more useful: *for problems where classical methods have encountered structural barriers, accepting an empirical constitution as the entry point can be a productive methodological choice.* This claim is independent of whether the problem is independent of ZFC. It applies whenever the constitutional structure is the productive formulation, regardless of the problem's status in formal logic.

6.3. **Methodological pluralism.** Mathematics benefits from multiple methodologies:

- Classical proof by contradiction: for statements provable from axioms.
- Constructive mathematics: for explicit object construction.
- Probabilistic methods: for deterministic results via probability (Erdős).
- Computational verification: for finite cases (four-colour theorem).
- *Spherical cow philosophy*: for structural barriers with identifiable empirical constitutions and cross-domain echoes.

These are not competing methodologies. They are tools for different situations. The contribution of this paper is to articulate the conditions under which the last tool is the right one, and to illustrate its application—including the ways it can go wrong—through the arithmetic programme of [20, 19, 22, 21].

7. Conclusion

7.1. **Summary.** Spherical cow philosophy provides a methodological complement to classical proof-by-contradiction for problems where structural barriers persist. By accepting empirical constitutional constraints as the productive entry point—deriving forced consequences rigorously and validating predictions across independent domains—the framework may open paths that a purely universal approach does not emphasise.

What the twin-prime case study shows:

(1) The constitution $p \equiv 2 \pmod 3$ is a theorem. It motivated the cascade moduli and the conjecture $\theta_W = 5/8$ [19], which is open; Condition W3 [22] is true but affects no level of distribution. None of these is forced by the constitution.

(2) The comparison with Shannon's AEP and Channel Coding Theorem [21] is a proposed analogy. In the AEP the distinguished subset is a vanishing fraction of the space, the opposite of the θ_k pattern.

(3) The Sophie Germain example is not a test of the framework: its local densities are the same as those of twin primes, and the value $1/2$ is predicted at both depths 1 and 2.
(4) The $k = 4$ prediction becomes a falsifiability test only once its terms are defined and fixed in advance.

7.2. **The philosophical shift.** From: mathematics begins with axioms and proceeds by deduction.

To: mathematics also begins with constitutional constraints and proceeds by rigorous derivation of their forced consequences.

This is not a rejection of rigour. The methodology stands or falls with the rigour of its derivations, and the case study shows that each claimed consequence must be checked as carefully as any proof. The shift is in what counts as the *starting point*, not in what counts as a *proof*.

7.3. **Final reflection.** Omar Khayyam is remembered for finding geometry in numbers: he classified the cubic equations and solved them by intersecting conic sections, and he took part in the calendar reform adopted in 1079 [18]. He did not find an arithmetic solution of the general cubic, and said so; the geometry of conic sections gave him a different entry point.

The spherical cow reveals what matters—provided one then checks that what was kept really forces what is claimed.

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